

Attitude adjustment: Trends in the semantics of clausal-embedding

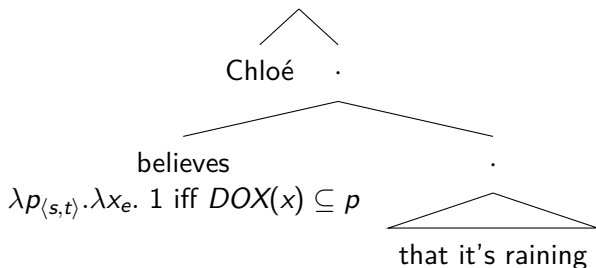
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Day 4

Some features of the system from yesterday

In sentences like 'x believes p'...

- the clause is the syntactic and semantic complement of 'believe'
- modality is introduced by 'believe'
- modality is anchored to the matrix subject



Back to nominals

We also talked about sentences like (1).

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- the proposition *that Nynke is going to be hired* is combining with that NP/entity.

Hintikka semantics is *tacit* about this, we need to refine it.

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- (2) Jort believes the rumor you told me **and** that Nynke is going to be hired.

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This argument rests on the “Law of the Coördination of Likes” and that (sic) there is no second silent ‘believe’ in the sentence.

Example (2) isn't a knock down argument for the single-verb view.

The LCL can be violated:

- (3) a. You can depend on my assistant and that he will be on time.
- b. *You can depend on that he will be on time.

The original example comes from Sag et al. (1985), but there has been a recent debate between Bruening & El Khalaf and Patejuk & Przepiórkowski about how to account for this.

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We know that there's ellipsis and there's the possibility of "conjunction reduction".

- (4) Jort believes the rumor you told me **and** <Jort believes> that Nynke is going to be hired.

Hirsch (2017) for a recent revival of conjunction reduction

The burden of proof rests on the person claiming that there are two verbs 'believe', and nobody afaik has written that paper yet.

Side note

Some among us might accept the thought that the syntactic category of sentence(-like) objects are different from the category of Noun Phrases.

(And I think historically, philosophers have believed that embedded clauses even in English were nominalized...)

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- (5) a. Ali'nin partner-in-i biliyorum.
Ali-GEN partner-3S.POSS-**ACC** I know
I know Ali's partner.
- b. [Ali'nin koş-tuğ-un-u] biliyorum.
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I know that Ali runs.
- (6) a. I know **that**.
b. I know **that** Ali runs.

The basic idea

Return to:

- (2) Jort believes the rumor you told me **and** that Nynke is going to be hired.

So, we would like a uniform semantics across *believe NP* and *believe CP* uses.

One way of doing so is...

Q1 to say that *believe* only combines with entities,

Q2 and to propose a way of combining NPs with clauses.

A proposal for Q1: Event semantics

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Subjects, objects and modifiers are introduced as conjuncts (in the semantics) and possibly by separate heads (in the syntax).

A proposal for Q1: Event semantics

Instead of saying, like we have been until now:

- (7) The sentence “Jones buttered the toast” is true at w iff
Jones buttered the toast at w .

we'll follow Davidson (1967) and his successors in writing:

- (8) $\llbracket \text{Jones buttered the toast} \rrbracket =$
 $\exists e : \text{butter}(e) \wedge \text{agent}(e) = \text{jones} \wedge \text{patient}(e) = \text{the toast}$

A proposal for Q1: Event semantics

- (9) $\llbracket \text{Jones buttered the toast} \rrbracket =$
 $\exists e : \text{butter}(e) \wedge \text{agent}(e) = \text{jones} \wedge \text{patient}(e) = \text{the toast}$

There are two things to note here:

- What in the heavens is e ? not the *type* e , the *variable* e !
- 'butter' is a transitive verb in the syntax, and yet its syntactic arguments are not being fed into the function that it denotes...

This all is okay.

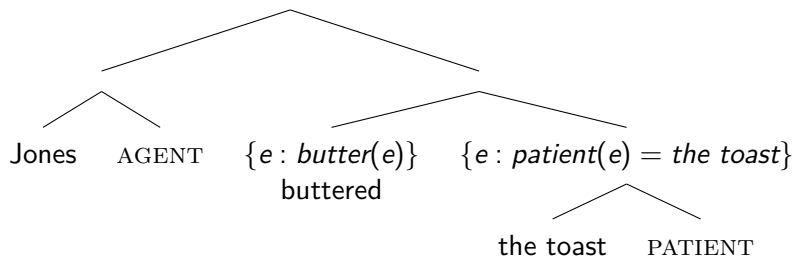
A proposal for Q1: Event semantics

One argument from Davidson

(10) Jones buttered the toast. **It** happened at midnight.

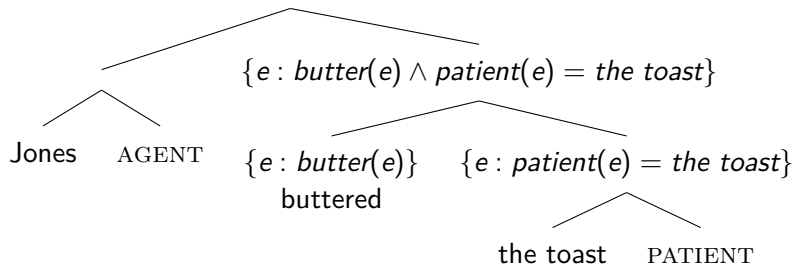
⇒ We can refer to events just like we refer to things: Events must be just like things.

One way of deriving these truth conditions

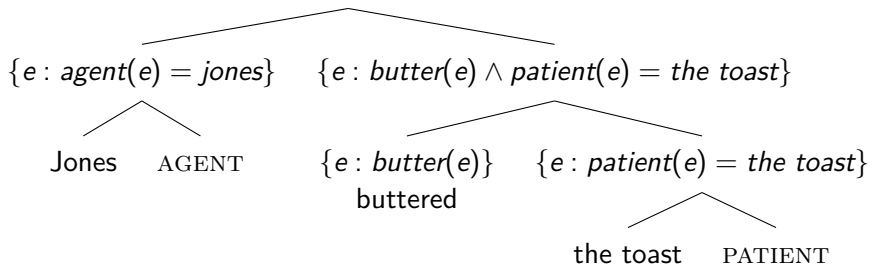


Expressions usually denote sets of events. Composition (here) proceeds via set intersection. There are multiple ways of implementing this.

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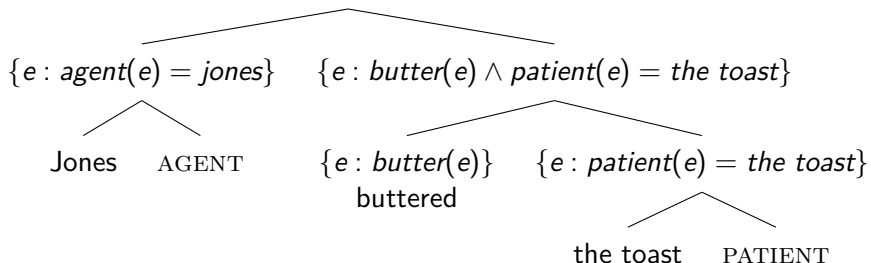


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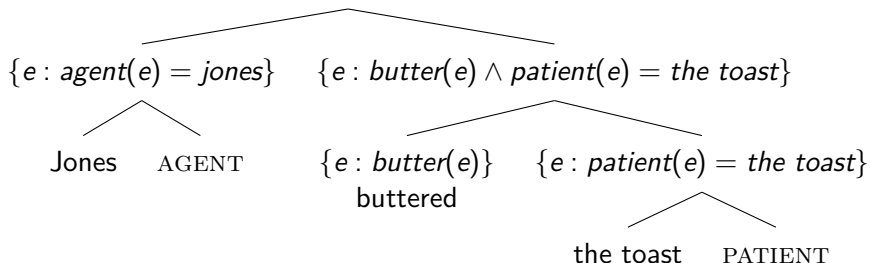
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The final step is to existentially quantify over this set:

$$(9) \quad \llbracket \text{Jones buttered the toast} \rrbracket = \\ \exists e : butter(e) \wedge agent(e) = jones \wedge patient(e) = the\ toast$$

A proposal for Q1: Event semantics

We now have:

- events,
- verbs 'severed' from their arguments (Kratzer 1996 for the term)

On to beliefs.

A proposal for Q1: Event semantics

On the same template as above, for the denotation of:

(11) Jort believes the rumor

we expect to get:

(12) $\exists e : \text{believe}(e) \wedge \text{experiencer}(e) = \text{jort} \wedge \text{theme}(e) =$
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Notes:

- If you worry about events vs. states, here we bundle the two together, but see Krifka et al. (1995).
- If you wanted to decompose *the rumor* further, you could (exercise on your own).

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This will now require us to update our semantics for *that* clauses.

Recall:

Q2 to propose a way of combining NPs with clauses.

A proposal for Q2: Contents of entities

We've seen Noun Phrases of the form, alone and as the complement of 'believe':

- (13) the rumor that Ebenezer and Balthasar are going to break
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One additional fact that people have observed are sentences like (14a), which feel like (14b):

- (14) a. The rumor **is** that Ebenezer and Balthasar are going to break up.
b. Clark Kent **is** Superman.

There's reason to think that this 'is' is the 'is' of equality. As opposed to the 'is' of predication:

(15) Clark Kent is fearless.

And I believe that this diagnosis is correct. See Potts (2007), Moulton (2009), or Bondarenko (2022) for discussion.

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(15) Clark Kent is fearless.

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There's a lot also to say about copular constructions...:

https://youtu.be/3MiPQ_edaFo?si=HoiUcKcJwIar4XP3

A proposal for Q2: Contents of entities

Based on these facts, we want to establish an equality between NPs like rumors, stories, facts, or indeed beliefs, and particular propositions...

(16) Informal:

- a. the rumor = that E and B are going to break up
- b. the belief = that it's raining

Kratzer (2006) and Hacquard (2006) are (we think) at the origins of the ideas presented here. See though work by Moulton, Bogal-Allbritten, Elliott and Bondarenko.

Disclaimer

As Setayesh also said this morning, some people believe in $\subseteq$ and others in $=$ here.

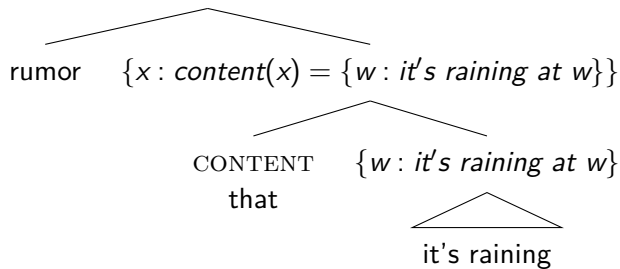
I actually don't think it follows from the copular clause facts that "the rumor that p " should involve $=$, but authors give other arguments.

A proposal for Q2: Contents of entities

One way to implement this is to use of of 'content functions'. We assume that certain NPs have content (rumors) and certain don't (you or me). Then we specify that content:

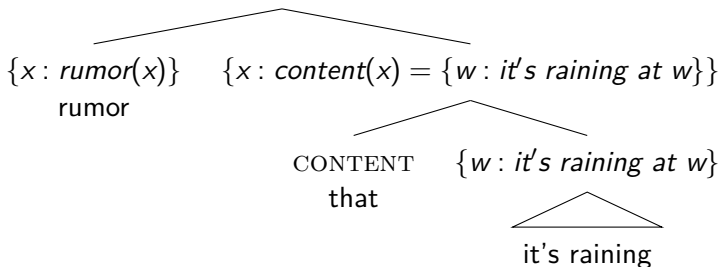
- (17) a. There's a rumor that it's raining.
b. $\exists x : \text{rumor}(x) \wedge \text{content}(x) = \{w : \text{it's raining at } w\}$

Doing this compositionally



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$$\{x : \text{rumor}(x) \wedge \text{content}(x) = \{w : \text{it's raining at } w\}\}$$



At this stage, one can

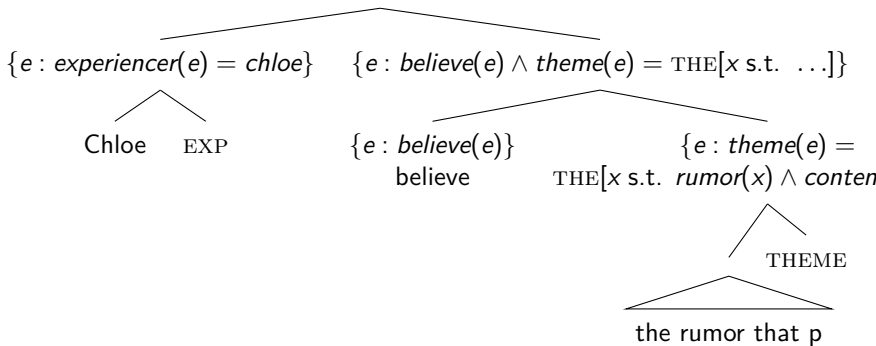
- existentially quantify over this set:
 $\exists x : \text{rumor}(x) \wedge \text{content}(x) = \{w : \text{it's raining at } w\}$
- turn it into a definite description:
 $\text{THE}[x \text{ s.t. } \text{rumor}(x) \wedge \text{content}(x) = \{w : \text{it's raining at } w\}]$

A Link to the Past

- rumors take us to possible worlds as well, just like beliefs,
- the function *content* functions just like the function DOX: it takes an individual and returns a set of worlds, ones compatible with the rumor in question.

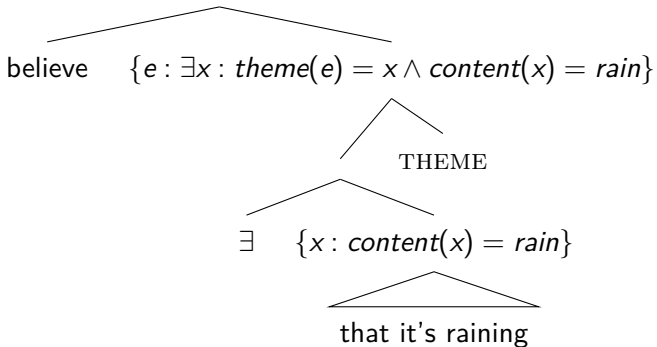
⇒ We've anchored a modal relation to an individual again, but that individual is not the *subject of 'believe'*. It's rather something that could serve as an NP *object of 'believe'*.

Doing 'believe the rumor that p' is fairly straightforward:

$$\{e : \text{believe}(e) \wedge \text{theme}(e) = \text{THE}[x \text{ s.t. } \dots]\} \wedge \text{experiencer}(e) = \text{chloe}$$


For 'believe that p', one needs to make an additional assumption
(given the system we've sketched out)...

$$\{e : \textit{believe}(e) \wedge \exists x : \textit{theme}(e) = x \wedge \textit{content}(x) = \textit{rain}\}$$



Adding in the experiencer and existentially closing over the final set of events, we get:

$$(18) \quad \exists e : \textit{believe}(e) \wedge \exists x : \textit{theme}(e) = x \wedge \textit{content}(x) = \textit{rain} \wedge \textit{experiencer}(e) = \textit{Chloe}$$

Given the contrast between a. and b.

- (19) a. Chloe believes the rumor that it's raining.
 b. Chloe knows the rumor that it's raining.

Discuss why the truth conditions in (18) might not be the perfect option for (19a).